

SRS — PERSONAL LOAN MANAGEMENT SYSTEM

1. Introduction

1.1 Purpose

The purpose of this SRS is to define the functional and non-functional requirements of the Personal Loan Management System.

It describes system behavior, UI flow, business rules, integrations, and processing logic for loan application, verification, approval, disbursement, and repayment.

1.2 Scope

The system enables customers to apply for personal loans online, upload documents, track status, repay EMIs, and close loans.

Backend users (Loan Officer, Admin) verify documents, decide eligibility, approve loans, issue sanction letters, and process disbursement.

The system includes:

- Customer portal
- Loan Officer portal
- Admin portal
- Eligibility engine
- Document verification
- Sanction and e-sign
- Disbursement
- EMI + Ledger system
- Overdue & default management

1.3 Users

- **Customer**
 - **Loan Officer**
 - **Admin**
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2. Overall Description

2.1 Product Perspective

A monolithic web application with a backend API and frontend UI.

The system integrates with:

- Mock CIBIL check
- Dummy payment gateway
- Basic file storage (local / S3)
- Notification system (SMS/email – mock)

2.2 Product Features

- Online loan application
 - Real-time eligibility check
 - Document upload & verification
 - Loan approval workflow
 - Sanction letter generation
 - E-sign agreement
 - Loan disbursement
 - EMI schedule & ledger
 - Payment handling
 - Overdue tracking & default marking
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3. System Features

3.1 Customer Module

3.1.1 Registration

- User registers using Name, Phone, Email, Aadhaar, PAN, Password
- OTP verification required
- System checks phone/email uniqueness
- Basic KYC saved

3.1.2 Login

- Login using email/phone + password
- JWT session handling

3.1.3 Dashboard

- Show profile summary
- Actions:
 - Apply New Loan
 - My Applications
 - Active Loan & EMI
 - Support

3.1.4 Apply for New Loan

Multi-step application (Stepper):

1. Personal Info
2. Income
3. Loan Details

4. Eligibility check
5. Document upload
6. Review & Submit

Eligibility Logic

- Age between 21–60
- Monthly income $\geq$ minimum threshold
- FOIR $\leq$ 40%
- CIBIL $\geq$ 700
- No active loan
- No existing default
- Apply only once every 3 months
- If rejected $\rightarrow$ apply after 6 months

Status after submit: APPLIED

3.1.5 Document Upload

Required:

- Aadhaar
- PAN
- Salary slip
- Bank statement (optional)
- Photo

Validation:

- Allowed formats: PDF/JPG/PNG
- Max size: 5MB

Status $\rightarrow$ **DOCUMENT_VERIFICATION_PENDING**

3.1.6 Track Application

User sees full status timeline:

- APPLIED
 - ELIGIBLE
 - DOCUMENT_VERIFICATION_PENDING
 - DOCUMENT_RETURNED_FOR_CORRECTION
 - DOCUMENT_APPROVED
 - LOAN_APPROVED
 - SANCTION_LETTER_SENT
 - SANCTION_LETTER_ACCEPTED
 - LOAN_DISBURSED
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3.1.7 Sanction Letter & Acceptance

- User views sanction letter PDF
 - Accept → goes to Agreement Signing
 - Reject → application ends
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3.1.8 E-Sign Agreement

- User draws/places signature
 - Agreement PDF generated
 - Status → **AGREEMENT_SIGNED**
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3.1.9 Loan Repayment UI

Customer can:

- View EMI schedule
- Pay EMI online

- View ledger
 - Download statements
 - View overdue alerts
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3.2 Loan Officer Module

3.2.1 Dashboard

- View assigned applications
- Filters: Pending Review, Verified, Returned, Rejected

3.2.2 Document Verification

Loan Officer checks:

- Aadhaar
- PAN
- Salary slips
- Bank statements
- Employment verification
- Fraud checks (duplicate PAN, existing defaults)

Actions:

- Approve → status: **DOCUMENT_APPROVED**
 - Return → status: **DOCUMENT_RETURNED_FOR_CORRECTION**
 - Reject → status: **DOCUMENT_REJECTED (FINAL)**
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3.3 Admin Module

3.3.1 Review Verified Applications

Admin sees only applications marked DOCUMENT_APPROVED by officer.

3.3.2 Loan Approval

Admin can approve or reject.

- If approved:
 - Status → **LOAN_APPROVED**
 - Sanction letter auto-generated
- If rejected:
 - Status → **LOAN_REJECTED (FINAL)**

3.3.3 Sanction Letter Management

- Generate PDF
- Send to customer
- Change status → **SANCTION_LETTER_SENT**

3.3.4 Disbursement

- After agreement signed
- Validate bank account
- Disburse loan
- Generate:
 - EMI schedule
 - Loan account
 - Outstanding balance

Status → **LOAN_DISBURSED**

3.4 EMI & Loan Account Module

3.4.1 EMI Schedule Generation

Uses standard EMI formula:

$$\text{EMI} = [P \times R \times (1+R)^N] / [(1+R)^N - 1]$$

Each month:

- Interest = outstanding × rate
 - Principal = EMI – interest
 - Outstanding updated
 - Ledger updated
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3.4.2 Loan Account Statuses

- ACTIVE
 - OVERDUE
 - DEFAULTED
 - CLOSED
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3.4.3 EMI Payment

- User pays via payment gateway (mock)
 - System updates:
 - Ledger
 - EMI status
 - Outstanding balance
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3.4.4 Overdue & Default Logic

- EMI missed → OVERDUE
- 90 days overdue → DEFAULTED
- Defaulted users blocked from future loans

- Alerts sent to loan officer
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4. Status Definitions (Complete Lifecycle)

APPLICATION STATUS

1. APPLIED
 2. ELIGIBLE
 3. NOT_ELIGIBLE (FINAL)
 4. DOCUMENT_VERIFICATION_PENDING
 5. DOCUMENT_RETURNED_FOR_CORRECTION
 6. DOCUMENT_REJECTED (FINAL)
 7. DOCUMENT_APPROVED
 8. LOAN_APPROVED
 9. LOAN_REJECTED (FINAL)
 10. SANCTION_LETTER_SENT
 11. SANCTION_LETTER_ACCEPTED
 12. SANCTION_LETTER_REJECTED (FINAL)
 13. LOAN_DISBURSED
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LOAN ACCOUNT STATUS

1. ACTIVE
 2. OVERDUE
 3. DEFAULTED
 4. CLOSED
-

5. Non-Functional Requirements

5.1 Security

- JWT auth
- Sensitive data encrypted
- Secure file upload
- Role-based access

5.2 Performance

- API response < 2 seconds
- Dashboard loads < 3 seconds

5.3 Reliability

- Automatic retries for external API calls
- Audit logs for all actions

5.4 Scalability

- Horizontally scalable
- Stateless API design

5.5 Usability

- Clear stepper-based application
- Mobile-friendly design
- Validation hints & error messages

6. System Integrations

6.1 CIBIL Check (Mock)

Returns score for eligibility.

6.2 Payment Gateway (Mock)

Handles EMI payments.

6.3 File Storage

Stores documents.

6.4 Notification System

Sends SMS/email for important events.